

PostgreSQL 16 → 17 Upgrade — Complete Steps with Actual Outputs

Phase 1: Install PostgreSQL 17 Packages

Step 1: Install libs first (dependency required)

```
rpm -ivh postgresql17-libs-17.10-2PGDG.rhel8.10.x86_64.rpm
```

Output:

```
warning: postgresql17-libs-17.10-2PGDG.rhel8.10.x86_64.rpm: Header V4
RSA/SHA256 Signature, key ID 08b40d20: NOKEY
Verifying... #####
[100%]
Preparing... #####
[100%]
Updating / installing...
 1:postgresql17-libs-17.10-2PGDG.rhe#####
[100%]
```

Step 2: Install main package

```
rpm -ivh postgresql17-17.10-2PGDG.rhel8.10.x86_64.rpm
```

Output:

```
warning: postgresql17-17.10-2PGDG.rhel8.10.x86_64.rpm: Header V4 RSA/SHA256
Signature, key ID 08b40d20: NOKEY
Verifying... #####
[100%]
Preparing... #####
[100%]
Updating / installing...
 1:postgresql17-17.10-2PGDG.rhel8.10#####
[100%]
```

Step 3: Install server package

```
rpm -ivh postgresql17-server-17.10-2PGDG.rhel8.10.x86_64.rpm
```

Output:

```
warning: postgresql17-server-17.10-2PGDG.rhel8.10.x86_64.rpm: Header V4
RSA/SHA256 Signature, key ID 08b40d20: NOKEY
Verifying... #####
[100%]
Preparing... #####
[100%]
Updating / installing...
 1:postgresql17-server-17.10-2PGDG.r#####
[100%]
```

Step 4: Install contrib package

```
rpm -ivh postgresql17-contrib-17.10-2PGDG.rhel8.10.x86_64.rpm
```

Output:

```
warning: postgresql17-contrib-17.10-2PGDG.rhel8.10.x86_64.rpm: Header V4
RSA/SHA256 Signature, key ID 08b40d20: NOKEY
Verifying... #####
[100%]
Preparing... #####
[100%]
Updating / installing...
 1:postgresql17-contrib-17.10-2PGDG.#####
[100%]
```

Step 5: Verify PostgreSQL 17 binary

```
/usr/pgsql-17/bin/postgres --version
```

Output:

```
postgres (PostgreSQL) 17.10
```

✓ PostgreSQL 17 binaries installed successfully.

Phase 2: Initialize PostgreSQL 17 Cluster

Step 6: Initialize new data directory

```
sudo /usr/pgsql-17/bin/postgresql-17-setup initdb
```

Output:

```
Initializing database ... OK
```

Step 7: Verify new data directory

```
ls -lh /var/lib/pgsql/17/data/
```

Output:

```
total 56K
drwx-----. 5 postgres postgres 33 Jun 14 15:45 base
drwx-----. 2 postgres postgres 4.0K Jun 14 15:45 global
drwx-----. 2 postgres postgres 6 Jun 14 15:45 log
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_commit_ts
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_dynshmem
-rw-----. 1 postgres postgres 5.4K Jun 14 15:45 pg_hba.conf
-rw-----. 1 postgres postgres 2.6K Jun 14 15:45 pg_ident.conf
drwx-----. 4 postgres postgres 68 Jun 14 15:45 pg_logical
drwx-----. 4 postgres postgres 36 Jun 14 15:45 pg_multixact
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_notify
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_replslot
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_serial
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_snapshots
drwx-----. 2 postgres postgres 25 Jun 14 15:45 pg_stat
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_stat_tmp
drwx-----. 2 postgres postgres 18 Jun 14 15:45 pg_subtrans
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_tblspc
drwx-----. 2 postgres postgres 6 Jun 14 15:45 pg_twophase
-rw-----. 1 postgres postgres 3 Jun 14 15:45 PG_VERSION
drwx-----. 4 postgres postgres 77 Jun 14 15:45 pg_wal
drwx-----. 2 postgres postgres 18 Jun 14 15:45 pg_xact
-rw-----. 1 postgres postgres 88 Jun 14 15:45 postgresql.auto.conf
-rw-----. 1 postgres postgres 31K Jun 14 15:45 postgresql.conf
```

✓ New cluster initialized successfully.

Phase 3: Configure PostgreSQL 17

Step 8: Check PostgreSQL 16 settings to carry over

```
grep -E
"^(listen_addresses|max_connections|shared_buffers|work_mem|wal_level|archive
_mode|archive_command|log_)" \
/var/lib/pgsql/16/data/postgresql.conf
```

Output:

```
max_connections = 100
listen_addresses = '*'
wal_level = replica
archive_mode = on
archive_command = 'cp %p /var/lib/pgsql/16/archive/%f'
shared_buffers = 128MB
log_destination = 'stderr'
log_directory = 'log'
log_filename = 'postgresql-%a.log'
log_rotation_age = 1d
log_rotation_size = 0
log_truncate_on_rotation = on
log_line_prefix = '%m [%p] '
log_timezone = 'America/New_York'
```

Step 9: Apply settings to PostgreSQL 17

```
sudo -u postgres vi /var/lib/pgsql/17/data/postgresql.conf
```

Set these values to match PostgreSQL 16 plus updated archive path:

conf

```
listen_addresses = '*'
max_connections = 100
shared_buffers = 128MB
wal_level = replica
archive_mode = on
archive_command = 'cp %p /var/lib/pgsql/17/archive/%f'
max_wal_senders = 10
archive_timeout = 60
log_destination = 'stderr'
log_directory = 'log'
log_filename = 'postgresql-%a.log'
log_rotation_age = 1d
log_rotation_size = 0
log_truncate_on_rotation = on
log_line_prefix = '%m [%p] '
log_timezone = 'America/New_York'
```

Step 10: Copy `pg_hba.conf` from PostgreSQL 16

```
cp /var/lib/pgsql/16/data/pg_hba.conf \  
  /var/lib/pgsql/17/data/pg_hba.conf
```

Output:

```
cp: overwrite '/var/lib/pgsql/17/data/pg_hba.conf'? y
```

Step 11: Create archive directory for PostgreSQL 17

```
mkdir -p /var/lib/pgsql/17/archive  
chown postgres:postgres /var/lib/pgsql/17/archive  
chmod 700 /var/lib/pgsql/17/archive
```

No output = success ✓

Phase 4: Stop PostgreSQL 16

Step 12: Stop PostgreSQL 16

```
sudo systemctl stop postgresql-16  
sudo systemctl status postgresql-16 --no-pager
```

Output:

- postgresql-16.service - PostgreSQL 16 database server
Loaded: loaded (/usr/lib/systemd/system/postgresql-16.service; disabled;
vendor preset: disabled)
Active: inactive (dead)
Docs: <https://www.postgresql.org/docs/16/static/>

```
Jun 14 15:47:28 pgserver systemd[1]: Stopping PostgreSQL 16 database  
server...  
Jun 14 15:47:30 pgserver systemd[1]: postgresql-16.service: Succeeded.  
Jun 14 15:47:30 pgserver systemd[1]: Stopped PostgreSQL 16 database server.
```

✓ PostgreSQL 16 is stopped.

Phase 5: Run `pg_upgrade` Check

Step 13: Change to postgres home directory

```
cd /var/lib/pgsql
```

⚠ This is required. `pg_upgrade` needs write access to current directory. `/var/lib/pgsql` is owned by postgres user.

Step 14: Run compatibility check

```
sudo -u postgres /usr/pgsql-17/bin/pg_upgrade \
--old-datadir /var/lib/pgsql/16/data \
--new-datadir /var/lib/pgsql/17/data \
--old-bindir /usr/pgsql-16/bin \
--new-bindir /usr/pgsql-17/bin \
--check
```

Output:

```
Performing Consistency Checks
```

```
-----
Checking cluster versions                                ok
Checking database user is the install user              ok
Checking database connection settings                   ok
Checking for prepared transactions                     ok
Checking for contrib/isn with bigint-passing mismatch  ok
Checking data type usage                                ok
Checking for not-null constraint inconsistencies        ok
Checking for presence of required libraries             ok
Checking database user is the install user              ok
Checking for prepared transactions                     ok
Checking for new cluster tablespace directories         ok
```

```
*Clusters are compatible*
```

✓ All checks passed. Clusters are compatible.

Phase 6: Perform the Upgrade

Step 15: Run `pg_upgrade` with `--link`

```
sudo -u postgres /usr/pgsql-17/bin/pg_upgrade \  
  --old-datadir /var/lib/pgsql/16/data \  
  --new-datadir /var/lib/pgsql/17/data \  
  --old-bindir /usr/pgsql-16/bin \  
  --new-bindir /usr/pgsql-17/bin \  
  --link
```

Output:

Performing Consistency Checks

```
-----  
Checking cluster versions ok  
Checking database user is the install user ok  
Checking database connection settings ok  
Checking for prepared transactions ok  
Checking for contrib/isn with bigint-passing mismatch ok  
Checking data type usage ok  
Checking for not-null constraint inconsistencies ok  
Creating dump of global objects ok  
Creating dump of database schemas ok  
Checking for presence of required libraries ok  
Checking database user is the install user ok  
Checking for prepared transactions ok  
Checking for new cluster tablespace directories ok
```

If pg_upgrade fails after this point, you must re-initdb the new cluster before continuing.

Performing Upgrade

```
-----  
Setting locale and encoding for new cluster ok  
Analyzing all rows in the new cluster ok  
Freezing all rows in the new cluster ok  
Deleting files from new pg_xact ok  
Copying old pg_xact to new server ok  
Setting oldest XID for new cluster ok  
Setting next transaction ID and epoch for new cluster ok  
Deleting files from new pg_multixact/offsets ok  
Copying old pg_multixact/offsets to new server ok  
Deleting files from new pg_multixact/members ok  
Copying old pg_multixact/members to new server ok  
Setting next multixact ID and offset for new cluster ok  
Resetting WAL archives ok  
Setting frozenxid and minmxid counters in new cluster ok  
Restoring global objects in the new cluster ok  
Restoring database schemas in the new cluster ok  
Adding ".old" suffix to old global/pg_control ok
```

If you want to start the old cluster, you will need to remove the ".old" suffix from /var/lib/pgsql/16/data/global/pg_control.old. Because "link" mode was used, the old cluster cannot be safely started once the new cluster has been started.

```
Linking user relation files ok
```

```
Setting next OID for new cluster          ok
Sync data directory to disk              ok
Creating script to delete old cluster     ok
Checking for extension updates           ok
```

Upgrade Complete

Optimizer statistics are not transferred by pg_upgrade.
Once you start the new server, consider running:
 /usr/pgsql-17/bin/vacuumdb --all --analyze-in-stages
Running this script will delete the old cluster's data files:
 ./delete_old_cluster.sh

✓ Upgrade completed successfully.

Phase 7: Start PostgreSQL 17

Step 16: Start PostgreSQL 17

```
sudo systemctl start postgresql-17
sudo systemctl status postgresql-17 --no-pager
```

Output:

```
• postgresql-17.service - PostgreSQL 17 database server
  Loaded: loaded (/usr/lib/systemd/system/postgresql-17.service; disabled;
vendor preset: disabled)
  Active: active (running) since Sun 2026-06-14 15:49:53 EDT; 67ms ago
  Docs: https://www.postgresql.org/docs/17/static/
  Process: 911544 ExecStartPre=/usr/pgsql-17/bin/postgresql-17-check-db-dir
${PGDATA} (code=exited, status=0/SUCCESS)
  Main PID: 911549 (postgres)
  Tasks: 8 (limit: 22960)
  Memory: 18.4M
  CGroup: /system.slice/postgresql-17.service
          └─911549 /usr/pgsql-17/bin/postgres -D /var/lib/pgsql/17/data/
             └─911551 postgres: logger
                └─911552 postgres: checkpointer
                   └─911553 postgres: background writer
                      └─911555 postgres: walwriter
                         └─911556 postgres: autovacuum launcher
                            └─911557 postgres: archiver
                               └─911558 postgres: logical replication launcher
```

```
Jun 14 15:49:53 pgserver systemd[1]: Starting PostgreSQL 17 database
server...
```

```
Jun 14 15:49:53 pgserver postgres[911549]: 2026-06-14 15:49:53.510 EDT
[911549] LOG:  redirecting log output to logging collector process
```

```
Jun 14 15:49:53 pgserver postgres[911549]: 2026-06-14 15:49:53.510 EDT
[911549] HINT: Future log output will appear in directory "log".
Jun 14 15:49:53 pgserver systemd[1]: Started PostgreSQL 17 database server.
```

✓ PostgreSQL 17 is running.

Phase 8: Post-Upgrade Verification

Step 17: Verify PostgreSQL version

```
sudo -u postgres psql -p 5432 -c "SELECT version();"
```

Output:

```
version
-----
PostgreSQL 17.10 on x86_64-pc-linux-gnu, compiled by gcc (GCC) 8.5.0
20210514 (Red Hat 8.5.0-28), 64-bit
(1 row)
```

✓ Running PostgreSQL 17.10

Step 18: Rebuild optimizer statistics

```
sudo -u postgres /usr/pgsql-17/bin/vacuumdb \
--all \
--analyze-in-stages \
-p 5432
```

Output:

```
vacuumdb: processing database "postgres": Generating minimal optimizer
statistics (1 target)
vacuumdb: processing database "template1": Generating minimal optimizer
statistics (1 target)
vacuumdb: processing database "test_replication": Generating minimal
optimizer statistics (1 target)
vacuumdb: processing database "postgres": Generating medium optimizer
statistics (10 targets)
vacuumdb: processing database "template1": Generating medium optimizer
statistics (10 targets)
```

```

vacuumdb: processing database "test_replication": Generating medium optimizer
statistics (10 targets)
vacuumdb: processing database "postgres": Generating default (full) optimizer
statistics
vacuumdb: processing database "template1": Generating default (full)
optimizer statistics
vacuumdb: processing database "test_replication": Generating default (full)
optimizer statistics

```

✓ Statistics rebuilt for all databases.

Step 19: Verify all databases

```
sudo -u postgres psql -p 5432 -c "\l"
```

Output:

```

                                List of databases
  Name          | Owner   | Encoding | Locale Provider | Collate  |
Ctype          | Locale | ICU Rules | Access privileges |          |
-----+-----+-----+-----+-----+-----+
 postgres      | postgres | UTF8     | libc             | en_US.UTF-8 |
 en_US.UTF-8   |         |          |                  |             |
 template0     | postgres | UTF8     | libc             | en_US.UTF-8 |
 en_US.UTF-8   |         |          | =c/postgres     |             |
               |         |          |                  |             |
               |         |          | postgres=CtC/postgres |             |
 template1     | postgres | UTF8     | libc             | en_US.UTF-8 |
 en_US.UTF-8   |         |          | postgres=CtC/postgres+ |             |
               |         |          |                  |             |
               |         |          | =c/postgres     |             |
 test_replication | postgres | UTF8     | libc             | en_US.UTF-8 |
 en_US.UTF-8   |         |          |                  |             |
(4 rows)

```

✓ All databases migrated successfully.

Step 20: Verify database sizes

```

sudo -u postgres psql -c "
SELECT
    pg_database.datname,

```

```
pg_size_pretty(pg_database_size(pg_database.datname)) AS size
FROM pg_database
ORDER BY pg_database_size(pg_database.datname) DESC;
"
```

Output:

datname	size
test_replication	2417 MB
postgres	1375 MB
template1	7798 kB
template0	7710 kB

(4 rows)

✓ Data sizes look correct.

Step 21: Verify tables

```
sudo -u postgres psql -p 5432 -d postgres -c "\dt"
```

Output:

List of relations			
Schema	Name	Type	Owner
public	big_table	table	postgres
public	test_replication	table	postgres

(2 rows)

✓ All tables present.

Step 22: Verify extensions

```
sudo -u postgres psql -p 5432 -c "
SELECT name, installed_version
FROM pg_available_extensions
```

```
WHERE installed_version IS NOT NULL
ORDER BY name;
"
```

Output:

```
   name   | installed_version
-----+-----
 plpgsql  | 1.0
(1 row)
```

✓ Extensions verified.

Step 23: Verify WAL archiving

```
sudo -u postgres psql -p 5432 -c "
SELECT archived_count, last_archived_wal, last_archived_time
FROM pg_stat_archiver;
"
```

Output:

```
 archived_count | last_archived_wal | last_archived_time
-----+-----+-----
              1 | 0000000100000002000000AB | 2026-06-14 15:49:24.859-04
```

✓ WAL archiving is active on PostgreSQL 17.

Phase 9: Enable Services

Step 24: Enable PostgreSQL 17 on boot

```
sudo systemctl enable postgresql-17
```

Output:

```
Created symlink /etc/systemd/system/multi-user.target.wants/postgresql-17.service → /usr/lib/systemd/system/postgresql-17.service.
```

Step 25: Disable PostgreSQL 16 on boot

```
sudo systemctl disable postgresql-16
```

No output = success ✓

Phase 10: Cleanup

Step 26: Delete old cluster data

Only after full verification:

```
./delete_old_cluster.sh
```

Or manually:

```
rm -rf /var/lib/pgsql/16/data
```

Upgrade Summary

Phase	Step	Action	Result
Install	1–5	Install PG17 RPMs	✓
Initialize	6–7	Init new cluster	✓
Configure	8–11	Copy settings, hba, archive	✓
Stop PG16	12	Stop old cluster	✓
Check	13–14	<code>pg_upgrade --check</code>	✓ Compatible
Upgrade	15	<code>pg_upgrade --link</code>	✓ Complete
Start PG17	16	Start new cluster	✓ Running
Verify	17–23	Version, DBs, tables, WAL	✓ All good
Enable	24–25	Enable PG17, disable PG16	✓
Cleanup	26	Remove old cluster	✓

Final Status

Item	Before	After
PostgreSQL version	16.x	17.10
Databases	4	4 ✓
Tables	present	present ✓
WAL archiving	active	active ✓
Archive count	581	1 (reset) ✓
Boot service	PG16	PG17 ✓

✓ Upgrade from PostgreSQL 16 to PostgreSQL 17 completed successfully.

Key fact

Detail

Method used	<code>pg_upgrade --link</code>
Downtime	minimal — only during upgrade execution
Data integrity	verified — all databases and tables present
WAL archiving	confirmed active on PostgreSQL 17
Old cluster	safely preserved until cleanup

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